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Integrating AI literacy into teacher education: a critical perspective paper

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Abstract
In today's educational landscape, equipping educators with AI literacy is crucial for creating equitable and effective learning environments. This perspective paper explores the challenges teachers face in developing AI literacy and advocates for training that goes beyond basic technical skills to include a deep understanding of AI mechanisms, applications, and ethical implications. Without this foundation, educators risk unintentionally deepening the digital divide, disadvantaging marginalized students. Using a literature-informed, narrative methodology, this paper integrates recent research and case studies, such as the U.S. E-rate program and India's "AI for All" initiative, as models for scalable solutions to promote AI equity. The paper introduces the EQUIP Framework (Ethical Governance, Qualified Professional Learning, Unified Collaborative Partnerships, Implementation Readiness, and Progressive Adaptation) to empower educators with the knowledge, skills, and ethical principles necessary for responsible AI use in education. Key considerations for effective implementation include customizing professional learning programs, strategically allocating resources, and establishing robust monitoring and evaluation processes. By addressing counterarguments related to resource constraints, ethical concerns, and risks of overreliance on technology, the paper offers a balanced perspective and provides practical recommendations. These emphasize the importance of integrating AI literacy into teacher education programs, ongoing professional learning, and ethical guidelines to enable educators to responsibly integrate AI, advancing a more inclusive and future-ready education system.

Keywords AI literacy, Teacher education, Professional learning, Digital divide, EQUIP framework, Ethical implications

1 Introduction

In today's rapidly evolving educational landscape, the integration of Artificial Intelligence (AI) presents both unprecedented opportunities and significant challenges. Many educators currently grapple with substantial obstacles in incorporating AI literacy into their teaching practices, primarily due to insufficient professional preparation and training. A recent global survey, which included over 450 schools and universities, revealed that fewer than 10% of them have established institutional policies or formal guidance regarding the use of generative AI applications [1]. AI literacy transcends basic

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technological understanding; it demands a nuanced comprehension of AI's inner workings, its myriad applications, and the ethical implications it presents [1, 2]. More critically, it requires the acumen to scrutinize AI outputs and make judicious decisions about their educational implementation. This multifaceted knowledge is the cornerstone upon which effective navigation and utilization of AI in learning environments is built.

For educators, AI literacy is not just about integrating cutting-edge tools into lesson plans. It's about equipping the next generation with the skills to thrive in an AI-dominated world. The stakes are high: without a robust foundation in AI literacy, educators risk becoming obsolete, potentially exacerbating the digital divide. The latter refers to the gap between those who have access to technology and internet connectivity—such as teachers and students—and those who do not [3], leaving vulnerable students at a significant disadvantage. Therefore, the imperative is clear: integrating AI literacy into teacher education programs is not just beneficial—it is critical for ensuring that educational technologies serve all students equitably, especially those from marginalized backgrounds.

This perspective paper advocates for the integration of AI literacy into teacher education programs to bridge gaps in educator preparedness and promote inclusive, equitable education. Using a literature-informed, narrative methodology, this paper integrates recent research and case studies to explore scalable solutions for promoting AI equity in education. Notable examples include the U.S. E-rate program, which addresses resource constraints, and India's "AI for All" initiative, which focuses on equitable AI access.

To tackle these challenges holistically, this paper presents the EQUIP Framework, which encompasses five key pillars: Ethical Governance, Qualified Professional Learning, Unified Collaborative Partnerships, Implementation Readiness, and Progressive Adaptation. This framework offers a structured approach to empower educators with the necessary knowledge, skills, and ethical principles for responsible AI use in education.

Additionally, this paper examines counterarguments related to financial constraints, ethical implications, and the risks of overreliance on technology, providing a balanced perspective on AI integration in education. It outlines practical strategies for implementation, such as tailored professional learning plans, strategic resource allocation, and cross-stakeholder collaboration, all aligned with the unique needs of diverse educational contexts. These strategies emphasize equitable access to resources and technology while fostering continuous professional growth and inclusive stakeholder engagement.

By addressing these key areas, the paper provides actionable recommendations for integrating AI literacy into teacher education. It underscores the importance of embedding AI literacy into initial teacher training, ongoing professional development, and ethical frameworks, empowering educators to responsibly integrate AI and build a future-ready, inclusive education system that benefits all learners.

The concept of "future-ready" education refers to equipping students with the skills, competencies, and adaptability needed to navigate an evolving technological and professional landscape. While the future remains uncertain, educational researchers and policymakers rely on emerging trends and forecasts to anticipate shifts in learning environments and workforce demands. Reports such as the EDUCAUSE Horizon Reports provide insights into the trajectory of educational technology, identifying AI-driven learning, automation, and digital fluency as key developments shaping the future of education [4]. The 2025 EDUCAUSE AI Landscape Study further highlights the growing

prioritization of AI in higher education, with 57% of institutions viewing AI as a strategic focus [5].

These projections highlight the urgent need to prepare students for an AI-integrated world, where digital literacy, critical thinking, and adaptability are essential. The rapid advancement of technologies—including robotics, virtual reality, blockchain, and quantum computing—further reinforces this need. Additionally, the COVID-19 pandemic accelerated the adoption of AI-powered learning platforms such as Zoom, Microsoft Teams, and Google Meet, demonstrating how education is increasingly shifting toward digital and hybrid learning models [6].

In light of these trends, future-ready education must emphasize not only technological proficiency but also critical engagement and adaptability in evolving digital landscapes. AI literacy involves a set of competencies that enable individuals to assess AI technologies critically, interact and collaborate effectively with AI systems, and utilize AI as a tool in various contexts, including online environments, workplaces, and daily life [7]. These skills are essential for navigating an increasingly AI-driven world, reinforcing the need for adaptability, digital fluency, and informed engagement with technology in education.

2 Methodology

This perspective paper adopts a literature-informed, narrative-driven approach to advocate for the integration of AI literacy in teacher education programs. This methodology synthesizes diverse scholarly and practical perspectives into a cohesive argument, examining the historical, current, and future implications of AI literacy for teacher education. A narrative-driven approach is particularly well-suited for this discussion, as it allows for a contextualized exploration of AI literacy beyond isolated empirical findings, integrating theoretical insights, policy analysis, and case studies to construct a well-rounded argument.

Rather than presenting a traditional systematic literature review, this paper interweaves findings from recent studies, empirical research, and policy reports to develop a nuanced discussion around AI literacy in education. Sources were selected based on the following inclusion criteria:

- Published from 2021 onward to ensure relevance to current AI developments,
- Peer-reviewed for academic rigor,
- Directly addressing AI literacy, teacher preparedness, ethical AI implementation, or educational equity,
- Representing both North American and international perspectives to provide a global view of AI literacy integration.

Exclusion criteria included non-peer-reviewed sources, industry-driven reports, and studies solely focused on technical AI model development, as these fall outside the scope of educational AI literacy.

To identify relevant sources, academic databases such as ERIC and Google Scholar were utilized. A snowball sampling approach was employed, allowing key studies to inform further exploration by reviewing their reference lists for additional relevant literature.

To analyze the selected literature, Braun and Clarke's six-phase framework for thematic analysis was applied [8]. This structured approach facilitated the identification,

analysis, and synthesis of recurring themes in the literature, such as barriers to AI literacy in teacher education, ethical concerns, and policy-driven solutions. Although typically used in qualitative research, thematic analysis was adapted here to systematically organize emerging themes across diverse sources, strengthening the paper's evidence-based advocacy.

All sources were assessed for methodological rigor, prioritizing clarity in research design, reliability, and relevance to teacher education. Preference was given to studies with transparent methodologies, replicable research designs, and clear data reporting. Special attention was also given to case studies and reports that highlighted practical strategies for AI literacy integration in diverse educational settings.

By adopting this methodological approach, the paper presents evidence-based recommendations to enhance educator preparedness, inclusive AI literacy initiatives, and ethical AI integration in teacher education.

3 Advancing inclusive and equitable education through AI literacy

3.1 Essential AI literacy for educators

Professional learning in AI literacy is both critical and urgent in today's rapidly evolving educational landscape. While AI offers promising benefits, such as personalized learning [9–11], these advantages may remain questionable if teachers' roles and competencies are not adequately addressed [2]. Therefore, insufficient preparation in AI literacy poses a significant barrier to its effective integration into K–12 curricula [12, 13].

A recent qualitative study on AI in English language teaching revealed that teachers often feel unprepared to use AI tools due to limited technological skills, with many citing a lack of training and support for integrating AI technologies into instructional practices [14]. Similarly, Su et al. [15] identified three key challenges: (1) insufficient AI knowledge, skills, and confidence among educators, (2) poorly designed curricula, and (3) the absence of teaching guidelines. Teachers are further constrained by overloaded curricula, inadequate infrastructure, and a lack of a clear framework for AI in Education [4, 16].

As AI technologies, such as intelligent agents, chatbots, and recommendation systems, become more prevalent in education, the urgency to address these gaps grows [15]. Teachers also face challenges in adopting AI tools, ranging from technical difficulties to communication barriers [16]. The potential pitfalls of AI—such as unreliable recommendations and algorithmic biases—highlight the need for educators to develop a nuanced understanding of AI's limitations and ethical implications [17].

Addressing these challenges requires targeted professional learning programs that not only build technical competencies but also cultivate ethical awareness and leadership for responsible AI use. Developing educators' digital competencies in AI literacy is essential to enabling them to critically assess, interact with, and effectively apply AI tools across diverse contexts [18]. However, as teachers transition from traditional knowledge providers to facilitators of AI-enhanced learning, they may experience tensions in redefining their professional roles [19]. These challenges are compounded by the influence of cultural contexts [20], which require educators to adapt AI-driven approaches in alignment with local educational norms, while also navigating shifts in professional identity [21]. Therefore, fostering AI literacy is crucial for educators, as it equips them to create learning environments where generative artificial intelligence (GAI) enhances educational

outcomes, promotes equity, enables personalized learning, and prepares students for future careers in an AI-driven world [22]. Given the pervasive societal impact of GAI, its integration into teacher education curricula is imperative [23–25]. Both preservice and in-service teachers must cultivate critical AI literacy skills to serve as role models for responsible AI use, thus preparing future generations to become "effective users, informed questioners, and reflective producers" (p. 172) [26]. While teachers need not become GAI experts, they must develop the necessary competencies and confidence to integrate AI in ways that are both responsible and innovative [16, 27].

If this critical need remains unaddressed, the transformative potential of AI in education risks being unrealized. Focusing solely on the practical applications of AI in education is insufficient. Teacher education programs must go beyond the basic use of AI tools to foster a deeper understanding of the technology—its functionality, its creators, the profits derived from it, and its broader societal implications [28, 29].

3.2 Personalized learning

AI literacy is essential for educators to implement personalized learning effectively, as it represents one of the key benefits of AI in education. Personalized learning tailors the educational process to each student's unique needs, skills, and pace [30–32]. By identifying problems and diagnosing them, tailored solutions can be developed to align with students' learning needs and preferences [33]. In this context, Education Technology provides a practical tool for the education sector to achieve personalization through data-driven approaches [34]. This approach empowers student agency by enabling learners to collaborate with teachers in setting goals, designing their learning experiences, and monitoring progress [35]. Personalized education focuses on forecasting students' performance and delivering targeted feedback to enhance learning, based on an analysis of their learning profiles and retention trends. Its primary goal is to refine the diagnosis, prediction, and improvement of learning outcomes while also working to prevent learning setbacks [34].

Machine learning (ML), a subset of artificial intelligence (AI), plays a pivotal role in delivering personalized learning experiences. By analyzing vast datasets, ML identifies patterns, predicts outcomes, and customizes educational pathways [36]. AI and ML play a vital role in collecting and interpreting learner data to enhance personalized learning. AI-driven learning platforms offer several benefits, such as an improved educational environment, adaptable scheduling, instant feedback, greater autonomy in managing learning experiences, and faster student progress [37]. These systems efficiently process extensive datasets, recognize patterns, and generate predictions or tailored recommendations. By considering individual strengths, abilities, and academic challenges, AI enables the application of diverse instructional approaches [38].

In education, ML processes student data to uncover trends, forecast academic performance, and recommend lessons that address individual needs [39]. For instance, these algorithms examine factors such as student performance, learning preferences, and behavior patterns to generate actionable insights. These insights shape the creation and delivery of personalized learning experiences by helping instructors identify individual strengths and weaknesses, anticipate learning requirements, and adapt lessons in real time [40].

AI enhances real-time data interpretation, providing educators with immediate insights into student participation, comprehension, and areas that require support [41]. This enables targeted interventions that close learning gaps and ensure progress by moving away from a one-size-fits-all model. Learners can advance at their own pace while receiving personalized support, fostering their academic success to its fullest potential [42]. AI-powered tutoring systems, which leverage natural language processing (NLP) and machine learning algorithms, assess student responses, monitor progress, and analyze learning behaviors. By doing so, these systems promote equitable access to education by reducing geographical and financial barriers, allowing students to learn at their own pace [42, 43]. Additionally, NLP can be used in chatbots to interact with students, providing feedback and personalized learning interventions [32].

In classrooms, AI supports differentiated instruction by enabling teachers to address diverse learner needs through personalization, enhancing the learning experience while improving teaching efficiency, allowing educators to allocate more time to innovative instruction and student engagement [44]. Moreover, smart content, such as digital books, video lectures, and virtual simulations, enriches learning experiences and streamlines teaching tasks, reducing teacher workloads and allowing for greater focus on direct instruction and student support [16].

However, achieving true equity requires addressing challenges such as access disparities, algorithmic biases, and privacy concerns, data security, as well as potential bias in algorithms and machine learning models [43]. Responsible use of adaptive AI algorithms is essential to mitigate these issues and realize their potential [41]. By harnessing AI thoughtfully and inclusively, educators and institutions can create learning environments that cater to the diverse needs of all students, particularly those from marginalized communities.

By recognizing individual differences, adapting instruction in real time, and fostering equity, AI has the potential to transform traditional education into a more inclusive, effective, and student-centered process. For this transformative approach to succeed, educators must be equipped not only to navigate the complexities of AI-driven tools but also to integrate them meaningfully into diverse learning environments. Teachers require a robust understanding of AI capabilities, ethical considerations, and practical strategies to align technology with pedagogical goals. As the adoption of AI in education accelerates, it is imperative to provide educators with targeted training and ongoing professional development. Without these efforts, the promise of AI-driven personalized learning may remain unfulfilled, preventing both students and teachers from fully benefiting from its transformative power. By prioritizing teacher preparedness, AI's full potential can be unlocked, creating equitable and student-centered learning environments.

3.3 Assistive technologies

AI-powered assistive technologies hold great potential to enhance accessibility for students with diverse needs. AI automates tasks that were once dependent on human input, such as speech and voice recognition, as well as decision-making processes [45]. AI-driven speech recognition systems, in particular, play a crucial role in supporting students with hearing impairments. These systems enable hearing-impaired students to engage more actively in classroom activities by instantly converting spoken language

into text. This transformation creates a more inclusive classroom environment, fostering engagement for both hearing-impaired students and their teachers [46]. Similarly, AI-based tools have been shown to have a positive effect in supporting student learning for mental disorders such as Attention-Deficit/Hyperactivity Disorder (ADHD), dyslexia, and autism spectrum disorder [47].

Moreover, initiatives such as Microsoft's collaboration with the Northwest Evaluation Association (NWEA) are improving accessibility in math assessments for students with vision disabilities, thereby creating opportunities for them to pursue advanced STEM careers [48]. Additional AI applications, including image recognition, facial recognition, lip-reading technology, and real-time captioning, provide critical support for students with sensory challenges or impaired communication disorders, ensuring their full engagement with learning resources across educational levels. For instance, AI-powered image captioning tools enhance the accessibility of visual content for students with visual impairments, while lip-reading technology aids individuals with hearing impairments by interpreting lip movements in real time. Real-time captioning technologies further enable students, including non-native speakers, to access spoken content in classrooms, promoting a more inclusive learning environment. These technologies are discussed in detail in a recent review that emphasizes their importance in supporting diverse learning needs through AI and assistive technologies [49]. Collectively, these technologies foster greater inclusivity, empowering students to succeed academically and pursue future careers in diverse fields.

AI also enhances English language teaching by offering personalized, adaptive, and data-driven support to students [8]. For example, intelligent tutoring systems analyze individual learner needs and proficiency levels to deliver tailored lessons and immediate feedback. AI-powered speech recognition tools help students improve pronunciation by analyzing speech patterns and providing instant corrections. Language learning apps powered by AI offer interactive lessons and personalized content recommendations based on student progress.

Additionally, AI can automate language assessments, such as essay grading. The emergence of AI tools like ChatGPT has demonstrated that machine learning systems can efficiently perform grading tasks and write reflectively, much like human markers [50–52]. While professional development helps teachers refine grading practices, AI systems can assist examiners by allowing them to focus on providing personalized feedback, which supports student progress more effectively [53]. Furthermore, NLP technologies enable AI chatbots and virtual tutors to engage students in conversation, facilitating language practice [54]. Text-to-speech and speech-to-text tools further enhance listening and reading skills by converting written text into speech, or vice versa. This multi-faceted approach illustrates how AI can be leveraged to support students in learning English more efficiently and effectively [55].

Moreover, virtual reality (VR) platforms, such as Oculus Rift, have incorporated AI into English as a Foreign Language (EFL) instruction. Foreign language learning refers to the study of a non-native language in a formal classroom setting, typically one not commonly spoken in the surrounding community [56]. In a study conducted by Harvard University, students engaged in immersive EFL lessons through AI-powered virtual environments. AI adapted the difficulty of exercises, provided real-time feedback on

pronunciation, and facilitated cultural experiences, creating an engaging and effective learning atmosphere [57].

4 Adressing counterarguments to AI literacy in teacher education

Despite the potential benefits of integrating AI literacy into teacher education programs, valid counterarguments raise significant concerns about feasibility, priorities, and the overall impact on education. This section examines key objections to AI literacy integration and proposes strategies to address and overcome these challenges.

4.1 Resources and financial constraints

Integrating AI literacy into teacher education programs is essential for promoting educational equity. However, implementing such programs requires substantial investments in technology infrastructure, including hardware (computers, servers, networking equipment) and software (AI tools and platforms). Technology has always been a central component of education, but the tools and platforms used today have evolved significantly, now playing a critical role in accessing digital learning resources and facilitating collaboration among students on online platforms. Ensuring equitable access to technology is essential for reducing disparities in education. Research indicates that access to the Internet, information and communication technologies, the quality and location of that access, as well as the types of devices available and the means to maintain them, significantly influence what individuals in well-connected countries, like the United States, are able or unable to do [58–61]. Therefore, students without access to devices such as laptops, tablets, or smartphones, as well as those lacking reliable internet, face significant challenges and substantial disadvantages. This gap impacts marginalized communities, further widening inequalities in academic success and future prospects, exacerbating the consequences of digital exclusion, and limiting the potential for all individuals to benefit from the opportunities AI provides [3, 62].

In addition, any underfunded institutions struggle to secure the necessary resources, which could further widen the digital divide and exacerbate existing educational inequalities [63]. The digital divide refers to disparities in access to digital devices, reliable internet connectivity, and essential digital literacy skills. These inequalities create substantial obstacles to quality education, disproportionately affecting students from low-income households, rural areas, and underserved communities. Socioeconomic status plays a critical role in shaping access to these digital resources, thus widening the gap in educational opportunities [64].

To address these challenges, collaboration among educators, policymakers, and community stakeholders is crucial. Such collaboration can help secure funding for professional development and implement policies to ensure equitable access. For example, the Federal Communications Commission's (FCC) Schools and Libraries Program (E-rate program) provides discounts of 20–90% on internet access, network equipment, and maintenance for eligible U.S. schools [65]. This initiative supports digital equity by ensuring access to high-speed internet and can serve as a model for AI literacy programs globally.

Equally important is the establishment of ongoing, school-based professional development programs that allow teachers to discuss context-specific challenges, share experiences, and explore effective AI integration strategies. Tailoring professional development

to the unique needs of each school enhances the relevance and sustainability of AI literacy initiatives.

4.2 Prioritizing transparency to address ethical implications

AI technologies in education raise important ethical concerns, including data privacy, algorithmic bias, and the potential for automation to reinforce stereotypes or discriminate against marginalized groups [66]. AI systems trained on unrepresentative datasets may perpetuate biases that disadvantage marginalized students. Frameworks, particularly those from organizations like the United Nations Children's Fund (UNICEF), emphasize the need to advocate for marginalized children and the importance of diversifying datasets to minimize these biases [67]. While AI engineers often strive to achieve fairness through technical solutions, such as creating unbiased algorithms and ensuring consistent service quality, the practical implementation of these methods remains uncertain [68]. Experts in AI ethics [69, 70] have argued that this technical approach to fairness is not always feasible or uniformly applied across different contexts.

Addressing these issues requires a commitment to transparency, accountability, and inclusivity in the design and use of AI systems [71].

Key actions to facilitate fairness include promoting transparency in algorithmic decision-making, safeguarding student data, and expanding access to technology to bridge the digital divide. This can help ensure that AI technologies benefit all students equitably. In this regard, initiatives like the One Laptop per Child (OLPC) Foundation illustrate how to provide children in developing countries with affordable laptops and digital learning resources [72]. By distributing rugged, low-cost laptops equipped with educational software, OLPC empowers children in underserved communities to explore, create, and connect through technology [73].

Additionally, the "AI for All" initiative by the Central Board of Secondary Education (CBSE) and Intel India promotes AI literacy through accessible, multilingual courses [74]. These efforts, alongside other targeted initiatives, may help promote transparency and inclusivity in AI education. However, further research and data are needed to fully assess their effectiveness in advancing ethical AI adoption.

4.3 Overreliance on technology

While AI technologies offer numerous benefits, overreliance on them in education presents significant risks. Overreliance on AI can occur when users accept AI-generated recommendations unquestioningly, which may lead to errors in task performance, particularly in decision-making. This often happens when individuals are unsure of how reliable the AI is or how much trust should be placed in its suggestions [75].

One major concern is deskilling, where the automation of tasks reduces human expertise, potentially leading to inefficiencies or disruptions if the technology fails. The rise of intelligent tutoring systems (ITS) and other AI-driven tools has been linked to the automation of teaching roles, reducing professional autonomy and potentially undermining the traditional expertise of educators [76]. To mitigate these risks, AI should be viewed as a complement to human interaction, not a replacement for it. Teachers should see AI as a tool that enhances their ability to provide personalized instruction and timely interventions, rather than as a substitute for their professional expertise [77, 78].

Incorporating AI into teacher education programs should emphasize its role as an adjunct to human-driven teaching strategies by involving all stakeholders, including teachers and students in the stages of its design [79]. This balanced approach allows educators to leverage AI's capabilities while maintaining focus on fostering sustainable solutions that respond to real-world issues [80].

Effectively addressing these concerns requires a multi-faceted approach that includes investment in infrastructure, ethical safeguards, and a balanced integration of AI tools. By proactively addressing these challenges, teacher education programs can prepare educators to use AI responsibly and equitably, contributing to a more inclusive and innovative educational landscape.

5 AI integration frameworks: comparative analysis and introduction of EQUIP

The integration of AI in education requires a structured and ethical approach to ensure its responsible and effective implementation. A recent UNESCO global survey [1] of over 450 schools and universities highlighted the significant gap in formal guidance regarding the use of generative AI applications in education. Specifically, 13% of universities indicated they have some form of guidance, while only 7% of schools have institutional policies in place. Additionally, nearly 40% of the institutions with guidance communicated it orally rather than through written documentation, underscoring the ad-hoc nature of the current responses. Around 20% of survey respondents were unsure whether their institution had any policies or guidance regarding AI, reflecting the uncertainty surrounding AI integration in educational systems. These statistics demonstrate the challenges educational institutions face in keeping up with the rapid emergence of AI technologies.

To effectively address the challenges of AI integration in teacher education, this paper argues for the development of a comprehensive and structured framework that guides the broader integration of AI into education. Such a framework can not only provide a clear, actionable roadmap for AI adoption across educational contexts but can also offer the necessary resources and guidelines to help ensure that teacher education programs are equipped to prepare educators for the ethical and effective use of AI in their classrooms. By establishing a solid foundation for AI integration in education, the framework can directly inform and support the integration of AI literacy into teacher training, facilitating educator preparedness to navigate and leverage AI technologies effectively.

5.1 Examining existing frameworks

As AI adoption in education continues to accelerate, various frameworks and models have been developed to support AI literacy and digital competency among educators. Examining these existing frameworks is essential to identifying best practices and addressing the need for a cohesive and comprehensive approach to integrating AI in education.

One prominent model is the UNESCO AI Competency Framework for Teachers and Students, launched in 2024 [81]. This initiative underscores UNESCO's commitment to empowering both educators and learners with the essential skills needed to navigate the evolving AI landscape in education. It aligns with Europe's digital competency frameworks, DigCompEdu and DigComp 2.2, which offer comprehensive structures for digital competency in the education sector. DigCompEdu [82] is designed to support the

development of educator-specific digital competencies in Europe, detailing 22 competencies organized into six key areas: professional engagement, digital resources, teaching and learning, assessment, empowering learners, and facilitating learners' digital competence.

Importantly, DigCompEdu focuses not on technical skills but on how digital technologies can enhance and innovate education. In contrast, DigComp 2.2 [83] presents the knowledge, skills, and attitudes needed for citizens to engage critically, confidently, and responsibly with digital technologies for learning, work, and societal participation. It identifies five key components of digital competence: information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving.

Another significant framework is the Framework for Inclusive AI Learning Design [84], which provides a comprehensive and adaptable approach for integrating AI into educational environments. Grounded in Universal Design for Learning (UDL) principles [85], this framework emphasizes inclusivity, accessibility, and equity, ensuring that AI-driven learning tools and curricula address the diverse needs of all students, particularly those from marginalized backgrounds. By aligning the "Five Big Ideas" of AI learning content with UDL's core principles—multiple means of engagement, representation, and expression—it advocates for flexible and personalized learning experiences that cater to various student abilities, preferences, and cultural contexts. Additionally, the framework stresses the importance of equipping educators with the knowledge and tools to use AI effectively, promoting ongoing professional development and collaboration. Real-world examples, such as a middle school AI summer camp, provide practical guidance, enhancing the framework's applicability in classroom settings. One of the major strengths of this framework is its foundation in UDL, ensuring that inclusivity and accessibility remain central to the design. This makes it a versatile and adaptable structure that can be applied across diverse educational contexts, from K-12 to higher education. However, the framework does not fully address other critical aspects of AI integration, such as data privacy, ethical considerations, and the technical infrastructure required for large-scale AI adoption in schools.

Furthermore, the AI Curriculum Design Framework [86] focuses on AI curriculum development in K-12 education. It aims to address the challenges educators face when developing and implementing AI-related curricula and is built around four key aspects: content, product, process, and praxis. The framework emphasizes three main content components—AI knowledge, AI processes, and the impact of AI—while integrating praxes such as student relevance, teacher-student communication, and flexibility. Grounded in data from interviews with 24 K-12 teachers across 12 schools, the model incorporates teacher perspectives to ensure practical relevance. However, this framework focuses on curriculum design and does not fully address critical challenges such as data privacy, algorithmic bias, or the infrastructure needed for large-scale AI adoption.

Moreover, AI Literacy Framework to Understand, Evaluate, and Use Emerging Technology [87], offers a concise and comprehensive guide for educational leaders to foster AI literacy in K-12 education. It integrates insights from research-based models, ethical decision-making benchmarks, and human justice perspectives to address the multifaceted challenges of AI in education. The framework emphasizes three interconnected "Modes of Engagement"—Understand, Evaluate, and Use—highlighting how these components work together to support robust engagement with AI. It underscores the

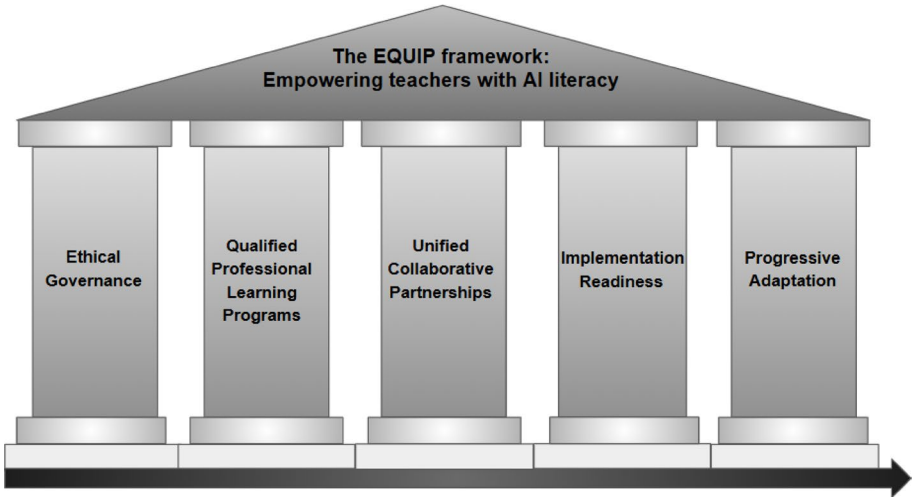

Fig. 1 The EQUIP framework: empowering teachers with AI literacy

importance of understanding how AI works, critically evaluating its societal and ethical implications, and using AI tools effectively and responsibly. Core values such as centering human judgment and justice are foundational to this framework, ensuring that AI applications in education promote equity.

In addition to these existing frameworks, the EQUIP Framework (Fig. 1), created by the author of this manuscript supported by literature, serves as a comprehensive guide to equipping educators with the necessary tools, skills, and ethical understanding for responsible AI integration in educational settings. It is structured around five interrelated pillars, each targeting a specific aspect of AI implementation.

5.2 Presenting the EQUIP framework

Figure 1 The diagram highlights the five key pillars of the EQUIP framework—Ethical Governance, Qualified Professional Learning, Unified Collaborative Partnerships, Implementation Readiness, and Progressive Adaptation—demonstrating their interconnectedness in fostering a holistic approach to AI literacy.

The first pillar of the EQUIP Framework is ethical governance, needed to ensure that AI integration in education prioritizes transparency, fairness, and accountability. This pillar advocates for the development of policies that uphold ethical standards, addressing issues such as bias, privacy, and the responsible use of AI. Although efforts have been made to establish AI policies, such as Singapore's AI governance and UNESCO's initiative [88], there remains a lack of consensus and conflicting advice regarding the use of AI in education and curricula [89]. The growing gap in AI integration is driven by unclear policies, limited awareness, variations in technical skills, and uncertainty about adoption [90].

Ethical governance fosters a culture of trust, aiming to create a secure, equitable, and inclusive educational environment where AI empowers, rather than excludes, students and educators. Additionally, it promotes AI ethics in education for students, encouraging an understanding of AI's social impact, biases, and ethical considerations, and preparing them to become informed, responsible citizens in an AI-driven world. The pillar also emphasizes community engagement, encouraging dialogue between schools and local communities about the societal implications of AI. Participation in global communities

of practice ensures that the ethical considerations of AI are continually refined through diverse perspectives, contributing to the development of global AI standards and best practices.

Qualified Professional Learning Programs are central to ensuring educators are equipped with the necessary AI knowledge and pedagogical skills for effective integration. This pillar emphasizes the creation of professional learning programs that:

- Establish foundational AI knowledge, ensuring teachers understand core concepts and the implications of AI. Teachers must continuously adapt to new technologies and develop essential skills to fully leverage AI in educational settings. They need to become tech-savvy by understanding and evaluating new ed-tech tools; they must also develop analytical skills to interpret data from these tools, enabling them to better understand learners and optimize their use of technology [91]. Additionally, refining teamwork, group management, and organizational skills is crucial for effectively integrating these tools into their daily teaching routines. Ethics training is an essential component of AI literacy programs, raising awareness of biases inherent in AI algorithms [92].
- Develop pedagogical and technological skills to enable educators to effectively integrate AI tools into their teaching strategies. This can be achieved by drawing on established frameworks such as TPACK (Technological Pedagogical Content Knowledge) [93] and the SAMR model (Substitution, Augmentation, Modification, and Redefinition) [94], a four-level taxonomy that outlines how technology impacts teaching and learning.
- Foster leadership and innovation by empowering educators to lead AI-driven initiatives and actively contribute to the evolution of educational practices.

Unified Collaborative Partnerships emphasize that AI integration in education is a shared responsibility. Therefore, this pillar focuses on fostering collaboration among key stakeholders such as schools, educational institutions, policymakers, and technology developers. AI literacy among all stakeholders is crucial, encompassing an understanding of AI capabilities, its application, and its ethical use [95]. This pillar aims to:

- Bridge resource gaps and ensure equitable access to AI technologies across diverse educational settings.
- Develop inclusive solutions that address the varied needs of learners and educators. Key concerns that have drawn global attention include AI's potential for discrimination and bias, threats to privacy, violations of human rights, and malicious uses of AI [96, 97]. The misuse of AI could lead to manipulation and social division, worsening inequalities [88].
- Promote a unified approach to AI in education, ensuring alignment between technological advancements and educational goals.
- Foster strong partnerships between school leaders and post-secondary institutions to enhance teacher preparedness programs. By bridging K-12 and higher education, these collaborations can ensure that pre-service and in-service teachers receive relevant, research-based training on AI integration, equipping them with the pedagogical and technological skills needed for effective classroom implementation.

This pillar also encourages the creation of global networks of educators, researchers, and policymakers who engage in shared learning to foster collaboration on AI solutions.

The Implementation Readiness pillar emphasizes preparing for the ethical and practical integration of AI into educational systems. Key priorities include:

- Promoting transparency in AI decision-making processes to build trust among stakeholders.
- Adopting human-centered AI collaboration ensures that technology enhances rather than replaces the role of educators. It emphasizes human agency by focusing on augmenting, rather than replacing, human abilities, promoting transparency in AI decision-making [98, 99], and enhancing the user experience by addressing human factors and social contexts.

The Progressive Adaptation pillar ensures that the EQUIP Framework remains alert to emerging challenges and opportunities as AI landscape evolves. Key aspects include:

- Continuous responsiveness to new ethical dilemmas and technological shifts, ensuring the framework adapts to the evolving nature of AI and education.
- Flexibility to accommodate diverse educational environments, ensuring the framework's relevance across varied educational landscapes.
- Active engagement in global conversations ensures that educators, researchers, and policymakers collaborate to address challenges such as equity, bias, and accessibility. This feedback loop keeps the framework dynamic and aligned with global trends in AI integration.

To critically evaluate the strengths and distinctions of the EQUIP framework, it is essential to examine how it compares to existing models guiding AI integration in education.

The UNESCO [1] and EQUIP frameworks both aim to enhance AI literacy in education, recognizing the need for teachers and students to develop foundational AI knowledge, critically evaluate its implications, and integrate it responsibly into educational settings. Structurally, the UNESCO framework organizes AI literacy into competency-based domains centered on understanding, application, and evaluation, whereas EQUIP adopts a systemic approach through its five interconnected pillars. While UNESCO emphasizes foundational AI literacy primarily for teachers and students, EQUIP expands on this by incorporating structured professional learning, leveraging established pedagogical models like TPACK, and fostering teacher leadership in AI-driven initiatives. Another key distinction is EQUIP's emphasis on collaborative partnerships between schools and post-secondary institutions to enhance pre-service teacher education and long-term AI integration strategies.

Similarly, the Framework for Inclusive AI Learning Design [84], grounded in UDL principles, prioritizes accessibility and equitable learning experiences. While EQUIP also values inclusivity, it operates at a broader systemic level, addressing not only classroom integration but also policy development, governance, and infrastructure readiness. The Inclusive AI Learning Design framework excels in flexible instructional design, yet it places a lesser emphasis on data privacy, AI ethics, and large-scale implementation challenges—areas that EQUIP directly tackles. Moreover, EQUIP's adaptive structure facilitates responsiveness to technological advancements and emerging ethical concerns, supporting its applicability across diverse educational contexts.

The AI Curriculum Design Framework [86], on the other hand, is tailored to K-12 AI education, centering on content, product, process, and praxis. It effectively supports curriculum planning by integrating key AI knowledge components, yet its primary focus remains on classroom instruction rather than system-wide AI governance and policy implementation. In contrast, EQUIP offers a comprehensive strategy, addressing ethical considerations, professional development, and infrastructure requirements that can ensure AI integration extends beyond curriculum design to institutional and policy levels.

Lastly, the AI Literacy Framework to Understand, Evaluate, and Use Emerging Technology [100], aligns with EQUIP in its commitment to AI literacy and ethical engagement. It focuses on three modes of engagement—Understand, Evaluate, and Use—providing structured guidance for educational leaders and students. While both frameworks emphasize ethical AI practices, EQUIP takes a broader, policy-driven approach, incorporating collaborative partnerships, implementation readiness, and governance models that can ensure AI adoption is equitable, transparent, and sustainable.

Ultimately, while each of these frameworks contributes valuable insights into AI integration, EQUIP distinguishes itself as a holistic, adaptable, and policy-driven model that can ensure long-term, ethical, and scalable AI implementation across educational systems.

While the EQUIP framework has not yet been formally tested, it is grounded in a robust theoretical foundation, built upon established principles of ethical governance, professional learning, and systems-based approaches to AI integration. The framework is designed to address gaps in existing AI integration models, such as long-term implementation readiness and scalability, making it a relevant and comprehensive tool for enhancing AI literacy and adoption across educational systems. Although empirical testing is still needed, the anticipated benefits of EQUIP can be inferred from its structure, which emphasizes ethical considerations, collaborative partnerships, and adaptability to evolving technological landscapes. Additionally, alignment with current trends in education underscore its potential impact. Future research can validate its effectiveness, but in the meantime, EQUIP provides a compelling framework that offers a holistic approach to AI integration in education, addressing the pressing need for sustainable and equitable AI practices.

5.3 Considerations and recommendations

Successfully integrating the EQUIP Framework into education systems requires a strategic and collaborative approach. The following recommendations outline key actions to promote ethical AI adoption, equitable access, and sustainable implementation.

- Adopt frameworks for annual policy reviews and public consultations, aligning with global standards to ensure compliance with international best practices.
- Redesign curricula to integrate AI ethics, embedding AI literacy into all disciplines with core modules on topics like algorithmic bias and data privacy.
- Develop written comprehensive guides for teachers on assessing and implementing AI tools, including ethical checklists and classroom strategies.
- Implement annual AI inclusivity audits to ensure tools are accessible to marginalized groups, including those with disabilities and non-native speakers. This may involve

partnering with third-party auditors to assess usage gaps in rural or underfunded schools and revise tool selection based on findings.
- Establish regional AI resource hubs to distribute free or low-cost AI resources and host workshops for educators in underserved areas.
- Collaborate with tech companies to provide infrastructure support for schools with limited connectivity.
- Develop an online portal for stakeholders to report AI-related concerns and track institutional responses publicly.
- Use feedback from audits, global networks, and public platforms to continuously refine AI integration strategies.
- Adopt a phased implementation approach, starting with pilot programs in select districts.
- Conduct longitudinal studies to measure outcomes in student performance, teacher efficacy, and equity metrics.

6 Conclusion

Integrating AI literacy into teacher education is not just a progressive step but an urgent necessity in today's rapidly evolving educational landscape. As AI becomes increasingly central to society, educators must be equipped with the skills and knowledge to harness its potential while navigating its challenges. Without a robust foundation in AI literacy, teachers risk becoming obsolete, exacerbating existing inequalities, and disadvantaging marginalized students.

This paper employed a narrative literature review to explore the integration of AI literacy in teacher education, synthesizing insights from literature, successful initiatives, and counterarguments such as financial and ethical concerns. The findings underscore the critical need for AI literacy training and offer practical strategies, informed by real-world examples, to overcome associated challenges. The proposed EQUIP Framework (Ethical Governance, Qualified Professional Learning, Unified Collaborative Partnerships, Implementation Readiness, and Progressive Adaptation), can provide a comprehensive, structured approach to empower educators with essential AI competencies and ethical principles.

The EQUIP Framework can serve as a roadmap to tackle barriers through targeted policies, interdisciplinary collaboration, and ongoing professional learning. By emphasizing ethical governance, the framework can ensure fairness and transparency in AI implementation. Qualified professional learning can equip educators with the skills to integrate AI effectively, while unified collaborative partnerships can foster stakeholder alignment. Implementation readiness can address logistical challenges, and progressive adaptation can ensure flexibility in response to emerging needs.

Prioritizing AI literacy through this framework can ensure that educators are not only adept at leveraging advanced tools but also prepared to create equitable and inclusive learning environments. The stakes are high, but the path forward is clear: Embedding AI literacy into teacher education programs, informed by the EQUIP Framework, could play a significant role in preparing educators for a future-ready education system. Empirical research is necessary to evaluate the practical implementation and impact of the proposed framework. Investigating its application in real-world contexts will provide insight into its effectiveness and allow for adjustments.

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R.D. independently conceptualized the study, conducted the literature review, and wrote the manuscript for this paper. All aspects of the study, including the design and interpretation of the argument, were carried out solely by this author.

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